JAY GANDHI

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Professional Summary

Senior AI Engineer with 4+ years of experience designing and delivering production-grade agentic AI, RAG, and backend software systems. Combines deep computer-science foundations with hands-on expertise in LLM orchestration, tool-integrated workflows, evaluation, and scalable API-driven architecture. Passionate about turning complex AI capabilities into reliable, measurable products that solve real-world problems.

Work Experience

Data Scientist, LTM (LTIMindtree - Banking and Financial Services Americas) – Pune, India 02/07/2026 – Present

- Built AI/ML capabilities for fraud detection and mitigation for a leading global banking client, supporting high-impact risk and financial-crime workflows.
- Developed an in-house prompt-validation tool that combines LLM-based assessment with human-in-the-loop review to improve prompt quality, consistency, and release confidence.
- Cleared the Karat Python assessment.

Research Engineer, Tata Consultancy Services – Pune, India 07/07/2022 – 29/06/2026

- Led applied research and innovation initiatives and designed and delivered data-driven AI solutions to solve large-scale enterprise problems in the space of legacy software modernization.
- Developed AI/ML pipelines to analyze enterprise scale structured and unstructured data (code, documentation, etc), generating actionable insights for stakeholders.
- Built a graph-based service boundary detection system using clustering algorithms on a weighted application knowledge graph with 50K+ edges, combined with LLM-assisted semantic labeling to improve decomposition accuracy.
- Designed an agentic AI workflow for automated monolith decomposition, orchestrating LLM calls with structured tool use and multi-step reasoning to reduce manual analysis time.
- Designed a RAG pipeline integrating LLMs with domain model documents for domain services to code traceability, reducing manual mapping effort by 70%; evaluated prompt designs to optimize retrieval quality.
- Built and exposed RESTful APIs for developed solutions, enabling seamless integration with front-end applications and enterprise systems.
- Delivered talks and demos at top tier internal/external forums; authored 3 ACM/IEEE papers and filed 1 patent.

Skills

Programming: Python, Java

AI/ML: LLMs, RAG, Agentic AI Workflows, Prompt Engineering, Fine-Tuning, Clustering, NLP, Embeddings

AI/ML Tools/Frameworks: LangChain, Hugging Face, Ollama, Pandas, NumPy, Google Agent Development Kit

DBMS: PostgreSQL, ChromaDB

Frameworks and Tools: FastAPI, Springboot, Docker, Git, Maven, Jenkins,

Practices: OOP, System Design, Microservices, REST APIs, Unit Testing

Education

Walchand Institute of Technology, Solapur, B.Tech (Computer Science and Engineering) CGPA: 9.62/10 2018 – 2022

Publications and Patents

Tracing Domain Services in Application Code using Generative AI (GenAI&SE, ISEC 2026) 2026

Microservices Identification using LLM (AISM, ASE 2025) 2025

Domain Aligned Microservices Decomposition (ISEC 2025) 2025

Method and system for identifying domain aligned microservices in enterprise applications (Patent) 2025

Certifications

Google Cloud Certified Generative AI Leader 2025
AWS Certified Cloud Practitioner 2023
Microsoft Certified Azure Fundamentals (AZ-900) 2022

Awards and Achievements

- Presented the work in-person at top tier (A*) international software engineering conference (ASE 2025)
- Presented the work in-person at Innovations in Software Engineering Conference (ISEC 2025, ISEC 2026)
- Received the IP Creation Award for filing a patent on LLM-based microservices decomposition.
- Finalist of TCS Global TACTICS 2024, an internal conference for technical architects.
- Won the Innovation Superstar Award for the publication of research papers.